

23,262

Developers Surveyed

8 Years

of Trend Data (2017-2024)

20+

Languages Tracked

State of Developer Ecosystem 2024

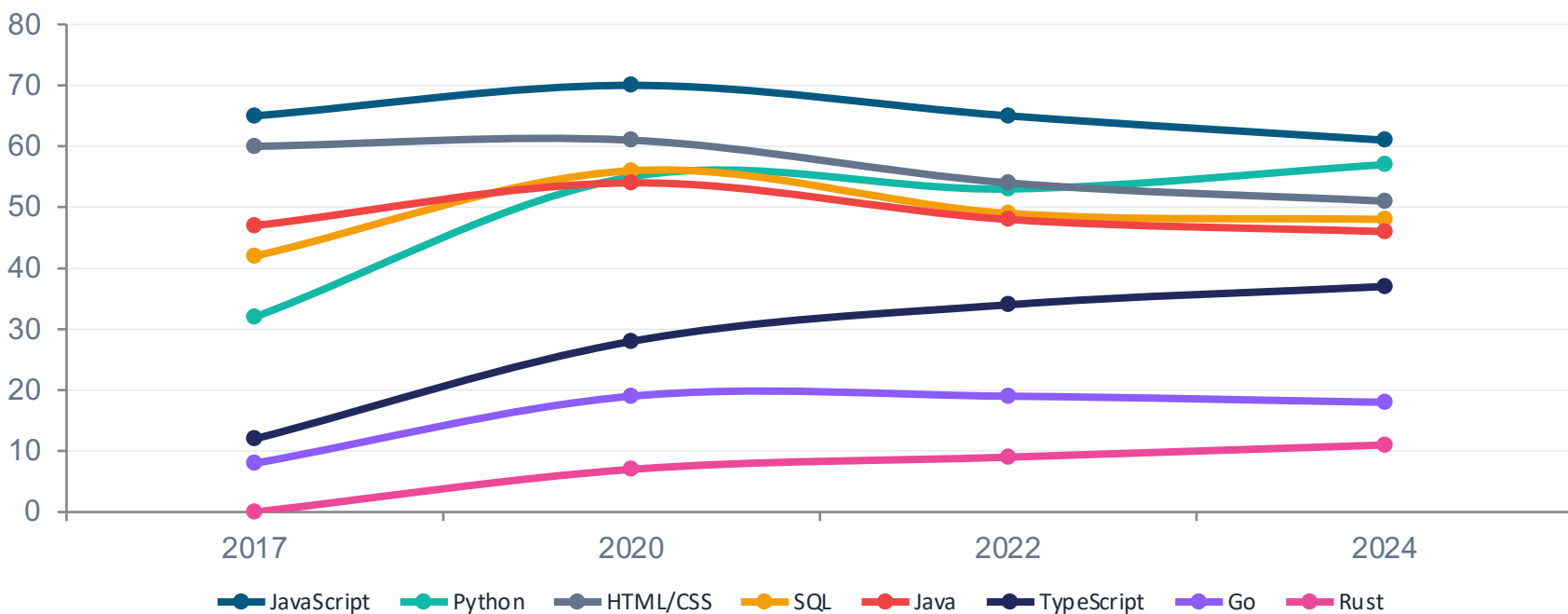
Language Trends, Satisfaction, and Strategic Shifts

JetBrains' annual survey captures the most comprehensive snapshot of the global developer community — covering programming languages, tooling preferences, AI attitudes, and career trends.

Source: JetBrains State of Developer Ecosystem Report 2024

Python Closes the Gap: 8 Years of Language Usage Shifts

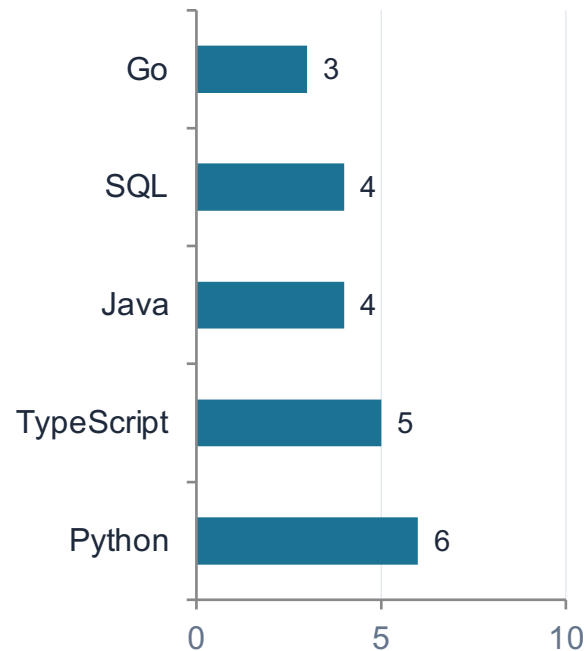
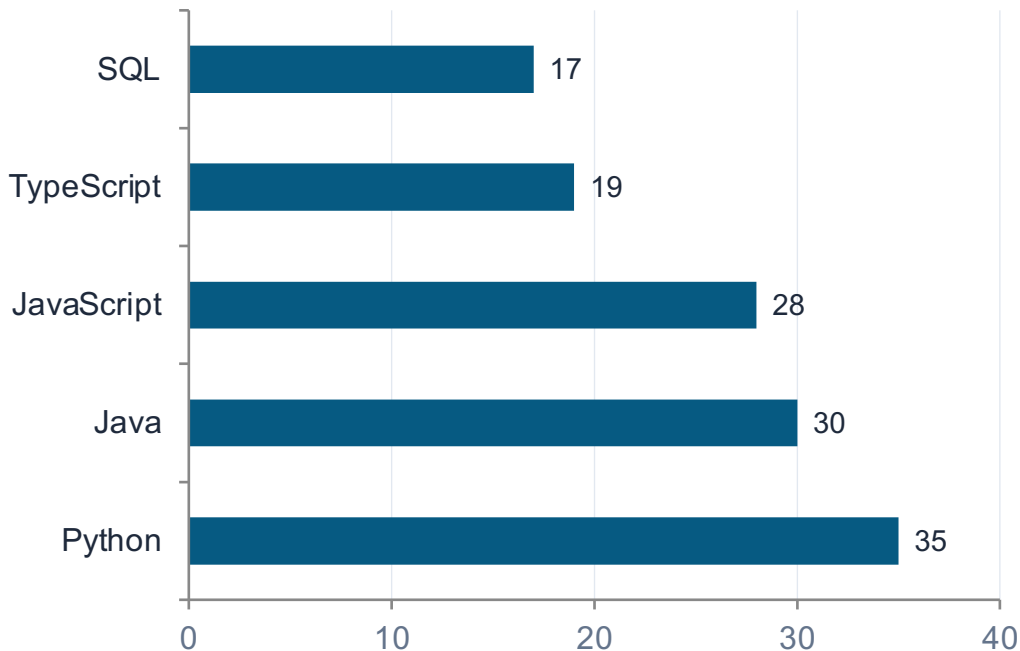
Usage share of top programming languages, 2017-2024 (% of developers)



Key Insight: Python rose from 32% to 57% while JavaScript declined from a 70% peak to 61%. TypeScript tripled from 12% to 37%.

Python Is Now the #1 Primary Language at 35%

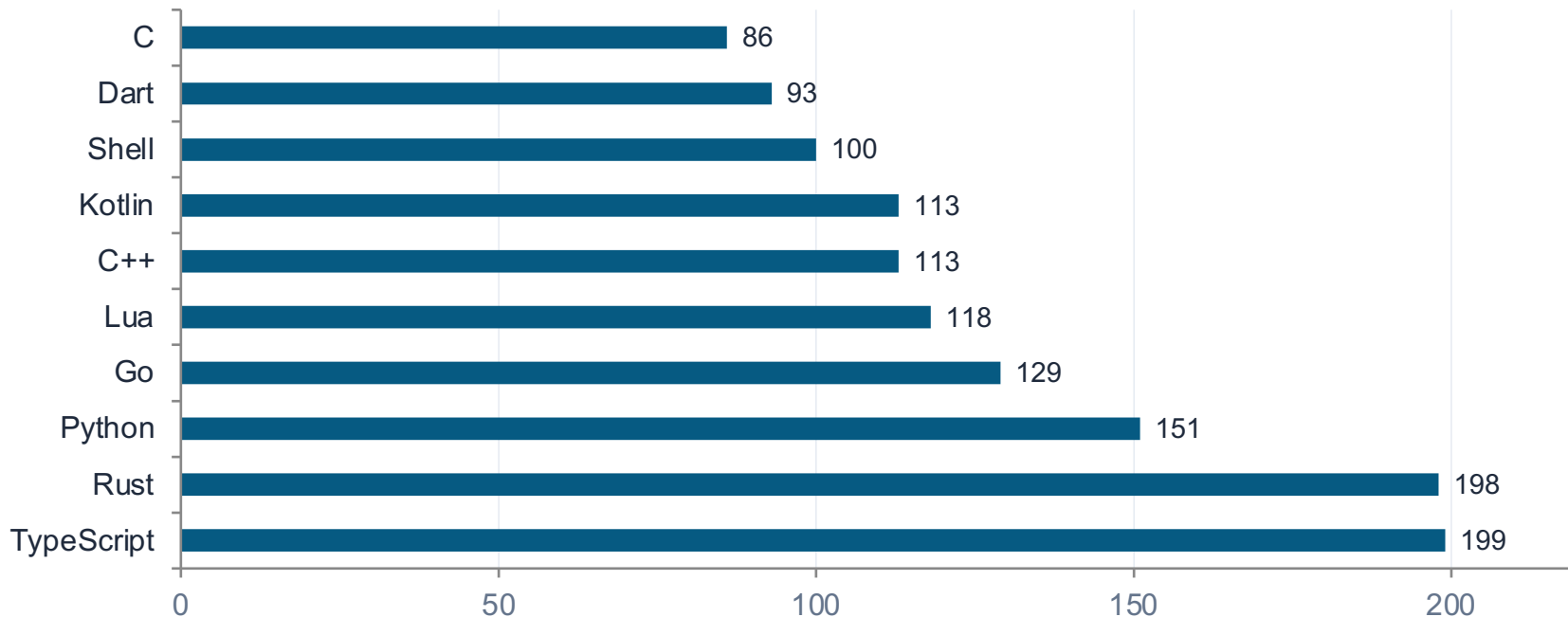
Primary language share vs. adoption intent (% of developers)



Insight: Adoption intent stays below 6% across all languages — developers rarely switch primary stacks.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Satisfaction scores reflect developer enthusiasm for type safety, modern tooling, and expressive type systems



Top 2 are type-safe languages with modern tooling. Python's #3 position (151) is driven by its AI/ML dominance.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven by application domain, not general preference

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Science / Analytics	Python	28%
User Interface Development	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

Hire for the domain, not just the language. Python dominates AI/ML; TypeScript owns the frontend.

What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML Teams

Python is the undisputed leader in AI (31%), data science (28%), and data processing. Build your AI talent pipeline around it.

02

TypeScript Is the Future of Frontend

Tripled usage to 37% and tops satisfaction at 199. It now dominates UI development (62%). Standardize on it for web and full-stack.

03

Rust and Go Are Infrastructure Growth Plays

Go stabilized at ~18%; Rust climbed to 11%. Both target cloud-native, microservices, and systems where performance is non-negotiable.

04

PHP and Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17%; Ruby fell from 10% to 4%. Plan succession or migration if these are still in your stack.

05

Satisfaction Correlates with Retention

TypeScript and Rust lead satisfaction thanks to type safety and modern tooling. Factor developer experience into your retention strategy.

Source: JetBrains State of Developer Ecosystem Report 2024 (n=23,262)